

No. of Functional Features Included in the Solution

Date	5 July 2026
Team ID	XXXXXX
Project Name	AI Learning Support Assistant
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Subject Selection	Allows users to select their study domain before submitting a query.	Streamlit UI	Done	High
2	Learning Problem Input	Accepts the student's learning problem through a text area.	Streamlit UI	Done	High
3	BiLSTM Emotion Prediction	Predicts the user's learning emotion using the trained BiLSTM model.	<code>bilstm_model.py</code>	Done	High
4	BERT Emotion Prediction	Predicts emotions using the BERT transformer model for comparison.	<code>bert_model.py</code>	Done	High
5	Mixed Emotion Detection	Identifies primary and secondary emotions with confidence scores.	<code>mixed_emotion.py</code>	Done	High

6	AI Learning Guidance	Generates personalized study guidance using Google Gemini API.	gemini_service.py	Done	High
7	Interaction History	Stores and displays previous learning interactions with prediction details.	interaction_logger.py, history_loader.py	Done	High
8	Analytics Dashboard	Displays emotion distribution, confidence trends, subject analytics, and summary statistics using Plotly.	Plotly Analytics	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	8

Additional / Nice-to-Have Features

Feature	Status
CSV Download of Interaction History	Done
Previous Interaction Viewer	Done
Session History Sidebar	Done
Clear History Function	Done
Responsive Streamlit Interface	Done
Emotion Probability Charts	Done
Analytics Summary Metrics	Done

Features Tested & Verified

- ✓ Subject selection
- ✓ Learning problem submission
- ✓ BiLSTM emotion prediction

- ✓ BERT emotion prediction
- ✓ Mixed emotion detection
- ✓ Gemini AI response generation
- ✓ Emotion probability visualization
- ✓ Interaction history storage and retrieval
- ✓ Previous interaction viewer
- ✓ CSV download functionality
- ✓ Analytics dashboard
- ✓ Clear history functionality
- ✓ Cross-browser compatibility
- ✓ Mobile responsiveness
- ✓ End-to-end application workflow

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Subject Selection, Learning Problem Input, Analytics Dashboard
2	Backend / Logic	3	BiLSTM Prediction, BERT Prediction, Mixed Emotion Detection
3	Database / Storage	1	CSV Interaction History and Session State
4	API / Integration	1	Google Gemini API Integration
5	Security / Authentication	0	Not implemented in the current project scope